

Presentation 4

Code link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/MenuTraffic_%26_HourlyTrends_29Oct_Krish.ipynb

Dashboard link:

https://app.powerbi.com/links/ZFnX_pLrbg?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbisource=linkShare&bookmarkGuid=d5a2ebd4-1ca7-47d6-a7d8-3ac2a01e0c6b

Presentation link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_20Nov_Katherine.pdf

What did you do?

I created a new Date table in the weekly trends and hourly trends legal menus with the code given to us, and linked it to the Date column in the main family_dash table. I also used the code given to us to make the Avg Daily Distinct CSID column in the main table. From this, I created the visualizations and I also double-checked that the seasonal trends graph had the correct code and columns.

How does it help the project?

Since we were having issues with accuracy last time, integrating this new and correct code will make sure that we are presenting the correct analysis to Cynthia.

Issues faced (if any):

I had issues with getting the code to work at first, which took awhile to resolve. I was also having issues with getting the data to display the way I wanted it to.

Attempts to resolve issues (if any):

I used ChatGPT and Claude to help me get the code to work and not throw errors, and I also realized there was a column selected that existed in a prior version of the table but didn't anymore, which was causing the problems with the visualization. I got rid of that column and fixed the visualization.

Issues resolved (if any)

All of the issues were resolved, and there is now a clear legal menu dashboard that Cynthia can click through for seasonal, weekly, and hourly trends.

Next steps:

I will conduct a comprehensive search for additional important information for the last presentation.

References (Mention if you built up on someone else's work):

N/A, did not use someone else's work for my analysis
Presentation 3

Code link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/CAR_Data_to_csv_Oct21_Katherine_Yang.ipynb

Dashboard link:

<https://app.powerbi.com/links/ecY92fxwuC?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbisource=linkShare&bookmarkGuid=b7ed9a64-c8f6-4af7-b12e-709720c9b2be>

Presentation link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_6Nov_Jasmine.pdf

What did you do?

I extended upon the drill down analysis that was presented to Cynthia to include the average number of calls by day of week, in addition to average number of calls by hours of day of the week.

How does it help the project?

This helps the project because we are able to see larger aggregates of data- hour of the day data is more variable, but with day of the week data, we can see broader trends in which queues people are accessing.

Issues faced (if any):

I had some issues understanding the Python code at first. I was also confused on how to create the DAX syntax for average calls per weekday.

Attempts to resolve issues (if any):

I used ChatGPT to break down the python code and understand what it was doing. Then I was able to export a csv file and upload it to Power BI. I also used ChatGPT to understand how to write the correct DAX formula for average calls per weekday.

Issues resolved (if any)

I can now click on any of the EP lines, and it will show me a drill down of the average number of calls there were for each day of the week, as well as average number of calls for each day of the week across next activity and the third activity.

Next steps:

We can also expand this drill down to include months, so we can give Cynthia a full drill down analysis spanning hours, days, and months.

References (Mention if you built up on someone else's work):

N/A, did not use someone else's work for my analysis
Presentation 2

Code link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/CAR_Data_to_csv_Oct21_Katherine_Yang.ipynb

Dashboard link:

<https://app.powerbi.com/groups/8cf04d4c-7aad-49fe-b493-02fbbdea9143/reports/e57e2915-edd9-4e01-ad2d-e4041c814061/0dc7c9724b403d251dd6?experience=power-bi&clientSideAuth=0>

Presentation link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_21Oct_Jasmine.pdf

What did you do?

I did a drill down analysis of the frequency of all calls across 6 different call lines (Benefits, Employment, Family, HIV, Housing, Immigration) by day of the week as well as by hours of the day.

How does it help the project?

This helps the project because we can see when client demands are highest, declining, and lowest throughout the week and the hours of each day, which helps Legal Aid allocate resources more efficiently and develop effective ways to manage overflow. I thought it was interesting how employment is the category with the least number of calls, and may ask Cynthia if there is a reason behind that.

Issues faced (if any):

I had some difficulty with creating the drill down analysis for days and hours of the day in PowerBI at first, but figured out how to do this using the hierarchy function. I also tried creating a drill down with Family SP and SP for all of the other lines, but encountered a lot of difficulties in doing so. I will try to resolve these issues during one of our work days.

Attempts to resolve issues (if any):

I was able to use Claude to help me successfully create the first drill down analysis. I found out that I was supposed to create a "hierarchy" on my day of the week column, and then add my Hours column to this hierarchy. Unfortunately it wasn't able to fix my issues for the second drill down analysis, so I will try to get that resolved in class.

Issues resolved (if any)

I can now click on any day of the week in PowerBI, and it will show me the number of calls there were for each EP line at each hour of the day.

Next steps:

I will analyze the quality of the calls (through call outcome, or creating a new column called missed call count) to see exactly how badly overflow is affecting their operations across these 6 lines. If overflow is a huge problem, we can try to create an automated phone message that informs users of the optimal times to call based on recent overflow patterns.

References (Mention if you built up on someone else's work):

N/A, did not use someone else's work for my analysis

Presentation 1

Code link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/tree/main/Code/Time%20Trends%20analysis

Dashboard link:

https://app.powerbi.com/links/1w9F1A5WU3?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pb_source=linkShare

Presentation link:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/Yang_Katherine_STAT390%20Presentation%201.pdf

What did you do?

I analyzed the frequency of all calls across 6 different call lines (Benefits, Employment, Family, HIV, Housing, Immigration) from 8am-8pm.

How does it help the project?

This helps the project because we can see when client demands are highest, declining, and lowest throughout the day, which can help Legal Aid allocate resources more efficiently. It was interesting to see that two lines in particular (Housing and Family) took up a disproportionate amount of the number of calls, which I want to explore more in the next part of the project.

Issues faced (if any):

I faced issues with trying to read the data into csv format at first, and trying to fully understand the meaning of all the columns in both datasets.

Attempts to resolve issues (if any):

I was able to use ChatGPT to help me successfully export a csv file containing the CAR data to my downloads folder. Additionally, reading the clarifications on the all calls data was very helpful to understand the columns.

Issues resolved (if any)

I loaded the complete CAR dataset into the free version of PowerBI and then was able to build my dashboard. I am still working on completely understanding the data, and which columns I want to use to construct my next analysis on.

Next Steps:

I want to analyze the quality of the calls (through columns like call outcome, or creating a new column called missed call count) to see exactly how badly overflow is affecting their operations, across these 6 different lines. 2 of the lines have extremely high frequency of calls, so I'd be

especially interested in understanding how overflow is affecting them. If overflow is a huge problem, we can focus specifically on solutions for that, such as creating a periodically updating note on their website that informs callers of the optimal times to call based on recent overflow patterns.

References (Mention if you built up on someone else's work): N/A, did not use someone else's work for my analysis